Our results suggest that a high *Lactobacillaceae* abundance prior to the onset of aGvHD may point to a preventive effect, as these patients survived. A human clinical trial of *Lactobacillus rhamnosus* GG prebiotic gavage to HSCT patients at time of engraftment demonstrated no protection against GvHD [1]. This could mean that *Lactobacillaceae* may not play a key role in aGvHD development, at least not the particular *Lactobacillus rhamnosus* strain under the conditions used in this group of patients. However, the administered probiotic did not alter the abundance of *Lactobacillus* spp. in the patients' guts [1], suggesting that the strain was not able to establish and proliferate in the host environment in this situation. An intrinsic increase of *Lactobacillaceae* prior to aGvHD onset, as observed here, therefore might still play a role in reducing aGvHD. A recent study related to the use of a probiotic given to infants to prevent sepsis suggested that the time point of application of a specific *Lactobacillus* sp. strain as a synbiotic played a critical role in positive clinical outcomes [2]. Furthermore, a study on gut microbial immunomodulation emphasized the importance of characterizing bacteria at the strain-level, because individual strains can have different modulatory effects on the immune system [3]. Therefore, it would be of great interest to determine the identity and predicted function of the specific *Lactobacillus* spp. strains in our patients, and in particular, in those who exhibited an early high abundance of *Lactobacillus* spp., as compared with those who experienced an expansion of *Lactobacillus* spp. after aGvHD and who later died.

Interestingly, *Enterococcus* was not among the most relevant taxa identified by our multivariate analyses. Intestinal domination of *Enterococcus* spp. was not clearly associated with adverse outcomes in our subgroup of 30 patients, in contrast to previous findings [4–6]. It should be noted that these previous observations were made in adult allo-HSCT patients and were dependent on the type and amount of antimicrobial treatment. In addition, to elucidate this discrepancy further, we are currently characterizing *Enterococcus* isolates from fecal samples of our patient group, to gain insight into bacterial strain-level differences.

Supplementary references

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